

Hospital Operations & Patient Analytics Dashboard

PROJECT DOCUMENTATION

Submitted as part of

Infosys Springboard Virtual Internship 7.0 – Batch 2

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PROJECT DETAILS

Project Information

Particulars	Details
Project Title	MedTrack_DV – Hospital Operations & Patient Analytics Dashboard
Project Domain	Healthcare Data Analytics
Internship Program	Infosys Springboard Virtual Internship 7.0 – Batch 2
Start Date	28 July 2026
End Date	17 September 2026
Student Name	Mythri L
Mentor Name	Prathina Likhita
Dataset Source	Kaggle
Programming Language	Python
Development Environment	Visual Studio Code, Jupyter Notebook
Data Storage and Processing	CSV, Microsoft Excel
Visualization Tools	Microsoft Power BI, Tableau

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1. PROJECT OVERVIEW

1.1 Introduction

Hospitals generate large amounts of data related to patient admissions, beds, wards, and hospital operations. Analyzing this information can help users understand how hospital resources are utilized and how patient-related activities change over time.

MedTrack_DV is a hospital operations and patient analytics project developed to convert hospital-related data into meaningful analytical insights.

The project begins with four datasets collected from Kaggle: admission.csv, bed.csv, patient.csv, and ward.csv. These datasets are processed sequentially using Python to create structured datasets suitable for analysis.

The processed data is used to calculate hospital performance indicators and prepare a final analytical dataset in Excel format.

Microsoft Power BI and Tableau are then used to develop interactive dashboards that present hospital information through charts, graphs, filters, and KPI visualizations.

1.2 Project Purpose

The purpose of this project is to simplify the analysis of hospital data by presenting important operational information in a visual format.

The dashboards help users explore patient admissions, hospital performance, department-related information, and resource utilization.

1.3 Project Deliverables

- Integrated and cleaned hospital datasets.
- Python scripts for data processing and KPI generation.
- Jupyter Notebook documenting data cleaning.
- Final analytical dataset in Excel format.
- Interactive Power BI dashboards.
- Interactive Tableau dashboards.

2. PROBLEM STATEMENT

2.1 Problem Statement

Hospitals maintain different types of information related to patients, admissions, beds, and wards. When this information is stored in separate datasets, analyzing hospital operations becomes more difficult.

Manually examining multiple datasets can make it challenging to understand admission patterns, compare hospital performance, and identify differences in resource utilization.

Another challenge is that raw datasets may contain missing values, duplicate records, inconsistent formats, or information that requires transformation before analysis. Therefore, a structured data analytics process is required to integrate the datasets, prepare the data, calculate meaningful performance indicators, and present the results in an understandable format. MedTrack_DV addresses this problem by using Python to process hospital datasets and Microsoft Power BI and Tableau to create interactive dashboards.

2.2 Proposed Solution

The project follows a sequential workflow:

1. Collect hospital datasets from Kaggle.
2. Integrate the datasets using Python.
3. Clean and transform the integrated data.
4. Generate hospital performance indicators.
5. Store the final analytical dataset in Excel format.
6. Develop dashboards using Power BI and Tableau.
7. Integrate and validate the completed dashboards.

2.3 Expected Benefit

The proposed system provides a structured and visual approach to hospital data analysis, making it easier to explore operational information and understand hospital performance.

3. PROJECT OBJECTIVES AND SCOPE

3.1 Objectives

The major objectives of MedTrack_DV are:

- To collect hospital-related datasets from Kaggle.
- To integrate admission, patient, bed, and ward information.
- To process the datasets using Python.
- To clean and transform the collected data.
- To generate hospital performance indicators.

- To prepare a final analytical dataset in Excel.
- To develop interactive dashboards using Power BI and Tableau.
- To present hospital-related information through visual analysis.
- To organize and validate the final dashboard deliverables.

3.2 Project Scope

The project focuses on hospital operations and patient analytics using the available dataset fields.

The analysis covers patient admissions, hospital performance indicators, department-related information, and resource utilization.

The project includes data collection, data preparation, KPI generation, dashboard development, integration, and validation.

3.3 Intended Users

The dashboards are designed to support the understanding of hospital operations by users such as:

- Hospital administrators.
- Healthcare managers.
- Data analysts.
- Students and researchers working with healthcare data.

3.4 Project Limitations

The accuracy and usefulness of the analysis depend on the quality and completeness of the collected datasets.

The project uses publicly available datasets and does not represent live hospital monitoring unless a live data source is specifically connected.

The dashboards are intended for analytical and educational purposes.

4. TOOLS AND TECHNOLOGIES

4.1 Kaggle

Kaggle is used as the source for collecting the hospital datasets required for the project. The admission, bed, patient, and ward files provide the initial data for analysis.

4.2 Python

Python is used to integrate, clean, transform, and prepare the hospital datasets. It is also used to generate the hospital performance indicators required for dashboard development.

4.3 Visual Studio Code

Visual Studio Code is used as the development environment for writing and executing the Python scripts used in the project.

4.4 Jupyter Notebook

Jupyter Notebook is used to document and perform the data cleaning and transformation process. It provides a structured environment for examining the dataset and recording the processing steps.

4.5 Microsoft Excel

Microsoft Excel is used to store the final analytical dataset in .xlsx format. The dataset serves as an input for dashboard development.

4.6 Microsoft Power BI

Power BI is used to create interactive dashboards that present hospital performance and patient-related information through visualizations, filters, and KPI displays.

4.7 Tableau

Tableau is used to develop additional interactive dashboards and explore hospital data through visual analysis.

5. PROJECT WORKFLOW AND METHODOLOGY

5.1 Project Workflow

The project follows a sequential data analytics methodology in which the output of one module becomes the input for the next module.

Data Collection → Data Integration → Data Cleaning → KPI Engineering → Dashboard Development → Integration → Testing

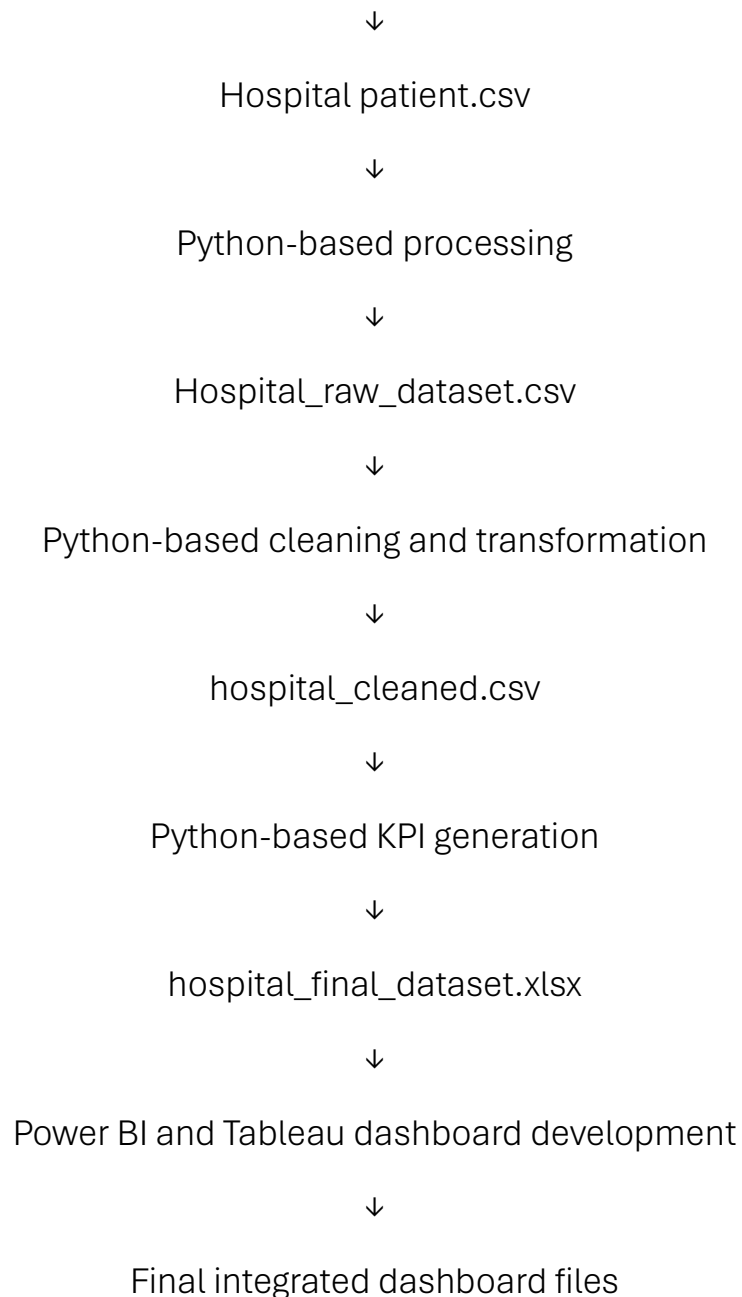
5.2 Sequential Dataset Transformation

The data transformation process begins with four separate CSV files.

admission.csv + bed.csv + patient.csv + ward.csv



Python-based integration



5.3 Methodology

The project uses a structured approach to transform raw healthcare data into a final analytical dataset.

Each module performs a specific task, beginning with dataset collection and ending with the integration and validation of the dashboards.

This approach helps maintain a clear relationship between the original datasets, the processed data, the calculated indicators, and the final visualizations.

6. MODULE 1 – DATA COLLECTION AND INTEGRATION

6.1 Objective

The objective of Module 1 is to collect the required hospital datasets from Kaggle and integrate them into a common dataset using Python.

6.2 Dataset Collection

Four CSV files are collected from Kaggle:

Dataset	Description
admission.csv	Contains hospital admission-related information
bed.csv	Contains bed-related information
patient.csv	Contains patient-related information
ward.csv	Contains ward-related information

These datasets provide different categories of hospital information that are required for the project analysis.

6.3 Data Integration

The collected datasets are processed using Python code according to the project procedure.

The integration process combines the required information from the four datasets into a single file named Hospital patient.csv.

The purpose of this stage is to bring the required hospital information together and prepare it for further processing.

6.4 Procedure

1. Download the four CSV files from Kaggle.
2. Organize the files in the project directory.
3. Load the datasets using Python.
4. Process the required information from each dataset.
5. Integrate the datasets using the implemented Python procedure.
6. Save the integrated output as Hospital patient.csv.

6.5 Output

Hospital patient.csv

6.6 Result

The four separate hospital datasets are integrated into a single file, providing the input for the next stage of data processing.

7. MODULE 2 – DATA CLEANING AND TRANSFORMATION

7.1 Objective

The objective of Module 2 is to process the integrated hospital data and prepare a structured dataset suitable for further analysis.

7.2 Input Dataset

Hospital patient.csv

7.3 Data Processing

The integrated dataset is processed again using Python code according to the implemented project procedure.

This stage transforms the integrated hospital data into the required raw dataset format. The generated output is saved as Hospital_raw_dataset.csv.

The dataset provides a consolidated representation of the hospital information collected in Module 1.

7.4 Data Cleaning

The project then uses Python-based cleaning and transformation procedures to prepare the data for analysis.

The cleaning process may include operations such as:

- Identifying duplicate records.
- Examining missing values.
- Standardizing inconsistent values.
- Preparing data fields for analysis.

The exact cleaning operations are determined by the Python code implemented in the project.

7.5 Output Files

- Hospital_raw_dataset.csv
- hospital_cleaned.csv
- hospital_cleaning.ipynb

7.6 Result

The integrated hospital data is transformed into a structured and cleaned dataset. This dataset becomes the input for KPI engineering in Module 3.

8. MODULE 3 – KPI ENGINEERING

8.1 Objective

The objective of Module 3 is to calculate hospital performance indicators and prepare the final analytical dataset.

8.2 Input Dataset

hospital_cleaned.csv

8.3 KPI Generation

The cleaned dataset is processed using Python code to generate the required hospital performance indicators.

The Python script performs the calculations and organizes the results into a format suitable for dashboard development.

The script used for this stage is named generate_hospital_kpis.py.

8.4 Final Dataset Generation

After KPI generation, the analytical results are stored in an Excel file named hospital_final_dataset.xlsx.

This file serves as the main dataset for developing the Power BI and Tableau dashboards.

8.5 Procedure

1. Load the cleaned hospital dataset.
2. Process the required data fields.
3. Calculate the hospital performance indicators.
4. Organize the analytical results.
5. Generate the final Excel dataset.

8.6 Output Files

- generate_hospital_kpis.py
- hospital_final_dataset.xlsx

8.7 Result

The cleaned hospital dataset is transformed into a final analytical dataset containing the information required for dashboard development.

9. HOSPITAL PERFORMANCE INDICATORS

9.1 Introduction

Key Performance Indicators (KPIs) are used to summarize important aspects of hospital operations.

The project plan identifies six hospital performance indicators for analysis.

KPI	Purpose
Total Admissions	Measures the number of hospital admissions
Occupancy Rate	Indicates the extent of hospital occupancy
Average Length of Stay	Measures the average duration of patient hospitalization
Readmission Rate	Indicates the frequency of patient readmissions
Bed Utilization Rate	Measures the utilization of available hospital beds
Department Efficiency Score	Supports comparison of department performance

9.2 Importance of KPIs

KPIs convert detailed hospital data into summarized information that is easier to understand.

They help users explore hospital performance, compare operational indicators, and identify patterns in the available data.

9.3 Use in Dashboard Development

The calculated KPIs are used in the dashboards to present important hospital information.

The visual representation of these indicators helps users understand the analytical results without examining the complete dataset manually.

The exact calculation formulas and displayed values depend on the Python implementation and the final analytical dataset.

10. MODULE 4 – POWER BI DASHBOARD DEVELOPMENT

10.1 Objective

The objective of Module 4 is to develop the initial hospital analytics dashboards using Microsoft Power BI.

10.2 Input Dataset

hospital_final_dataset.xlsx

10.3 Dashboard Development Procedure

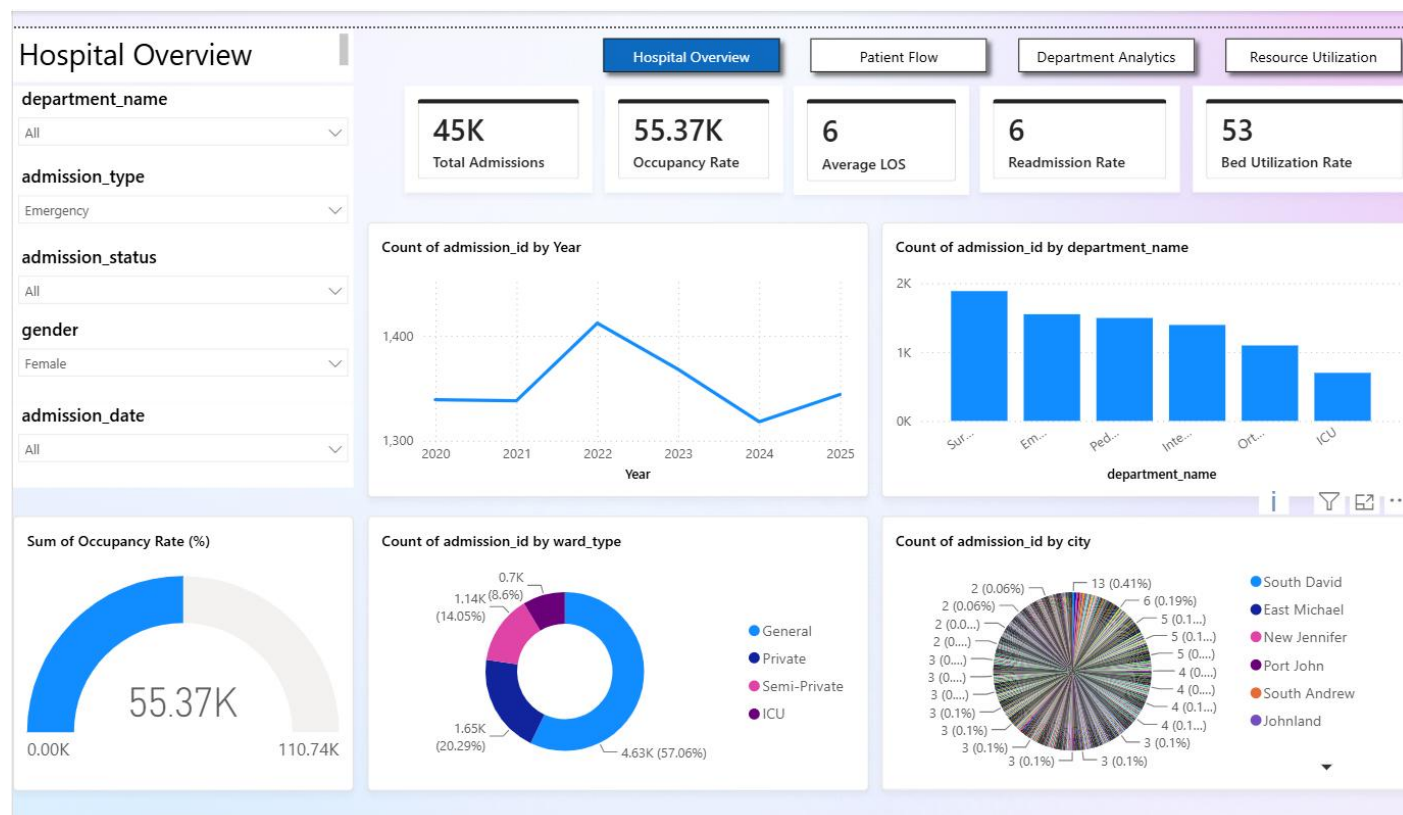
1. Import the final Excel dataset into Power BI.
2. Prepare the required data fields for visualization.
3. Create the initial dashboard pages.
4. Add the required charts and KPI visualizations.
5. Configure the available filters and navigation.
6. Review the dashboard layout and displayed information.

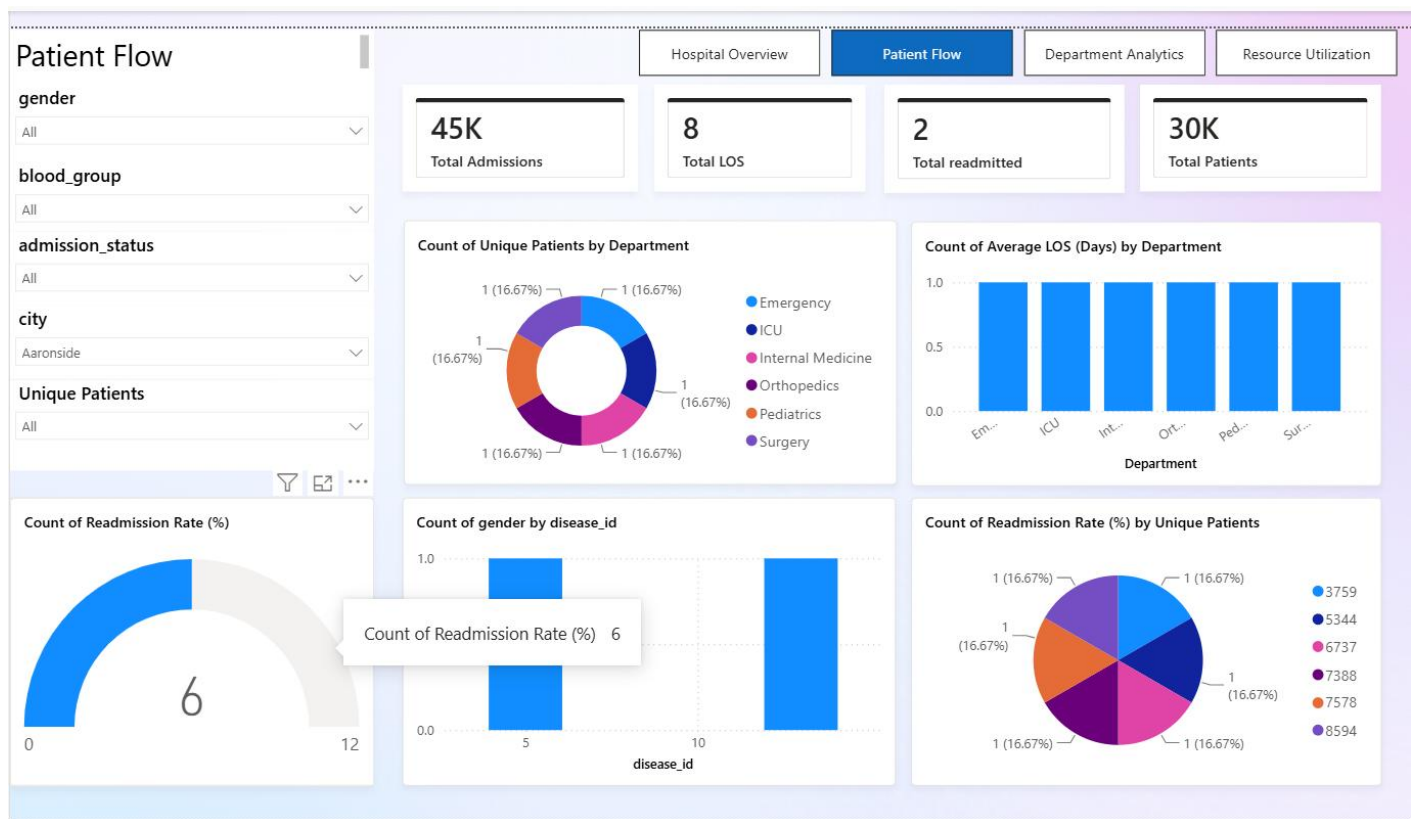
10.4 Dashboard Development

Power BI is used to present the hospital analytical data through visualizations.

The dashboard pages provide a structured view of the data and allow users to explore the information using the available interactive features.

10.5 Output





10.6 Result

The initial Power BI dashboards provide a visual representation of the hospital data and establish the foundation for the remaining dashboard development.

11. POWER BI DASHBOARD ANALYSIS

11.1 Dashboard Overview

The Power BI dashboards provide an interactive visual representation of the hospital analytical dataset.

The dashboards allow users to explore hospital information through charts, KPI displays, filters, and navigation features.

11.2 Dashboard 1

Dashboard title: Insert the exact title shown in Power BI.

This dashboard presents the hospital information available in the first dashboard page.

It can be used to examine the displayed hospital indicators and understand the overall patterns represented by the visualizations.

11.3 Dashboard 2

Dashboard title: Insert the exact title shown in Power BI.

This dashboard provides another view of the hospital analytical data.

The displayed visualizations help users explore the information represented in the second dashboard page.

11.4 Interactive Features

The dashboards include interactive features such as filters and navigation, where implemented.

These features allow users to explore the displayed information and move between dashboard pages.

12. MODULE 5 – TABLEAU DASHBOARD DEVELOPMENT

12.1 Objective

The objective of Module 5 is to extend the dashboard development using Tableau while retaining the first two Power BI dashboards.

12.2 Development Approach

The project retains Power BI Dashboard 1 and Dashboard 2.

Two additional dashboards are developed using Tableau to expand the visualization of the hospital analytical data.

12.3 Tableau Development Procedure

1. Connect the final analytical dataset to Tableau.
2. Prepare the required fields for visualization.
3. Create the first two Tableau dashboards.
4. Add the required charts and visual elements.
5. Configure the available filters and dashboard interactions.
6. Review the dashboards for consistency with the project objectives.

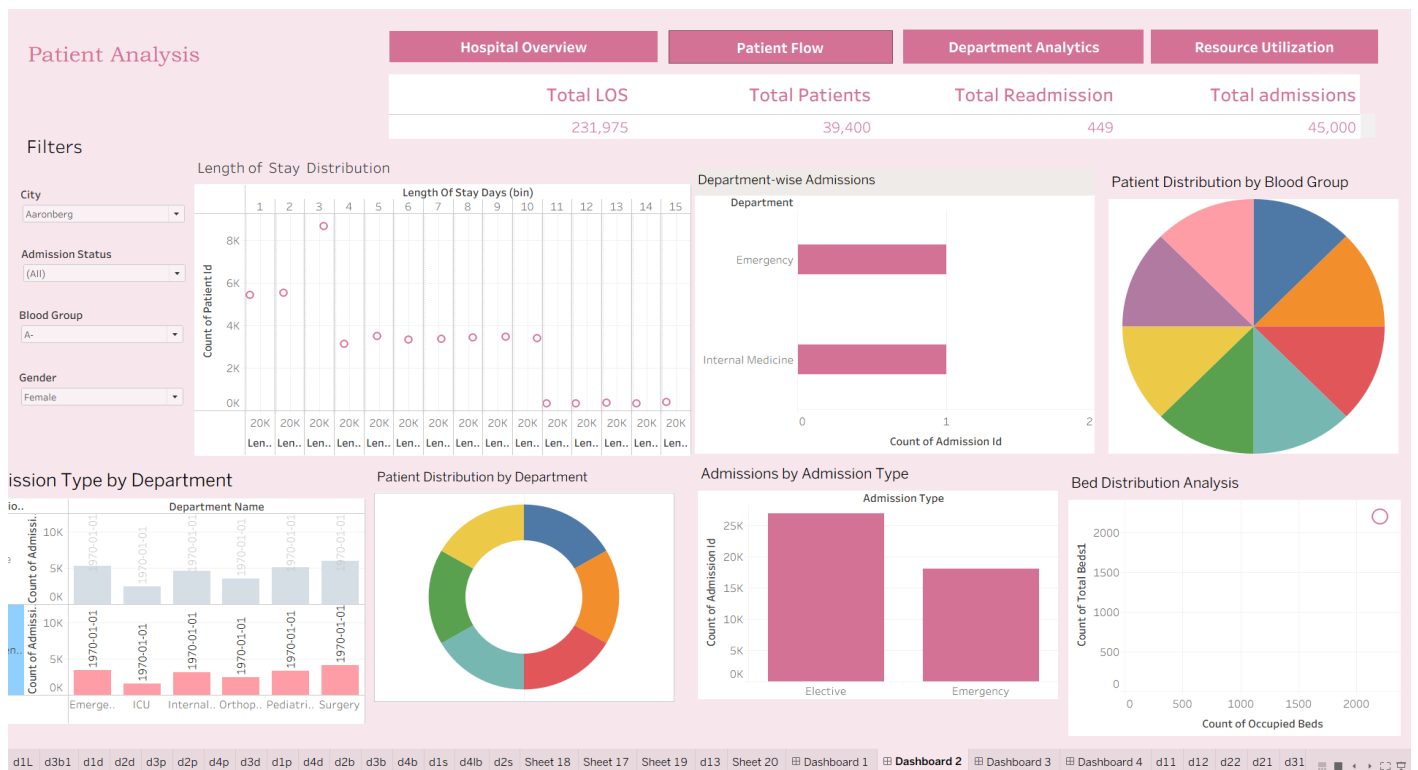
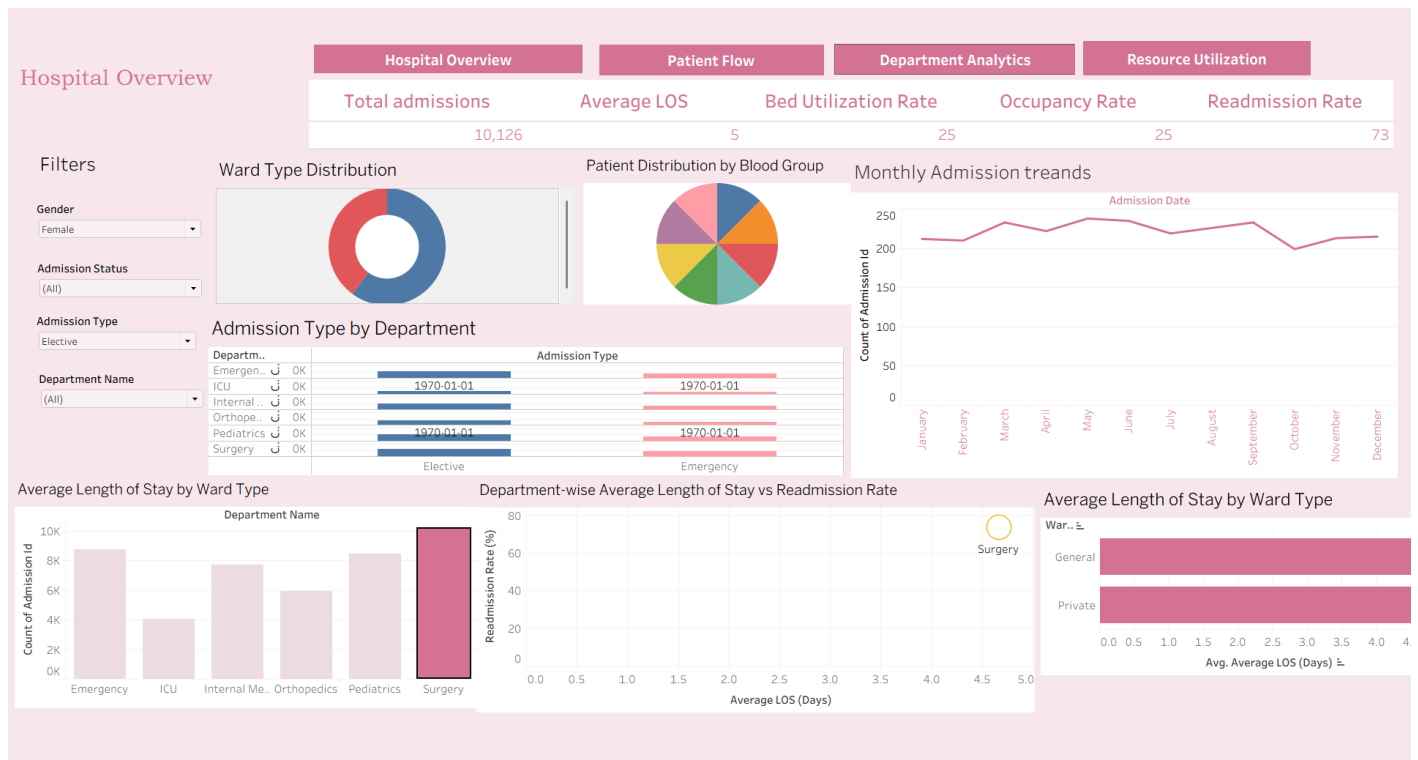
12.4 Dashboard Development

Tableau is used to create additional visual representations of the hospital data.

The dashboards provide another environment for exploring the available information and presenting the analytical results.

12.5 Result

The first two Tableau dashboards are developed and combined with the existing Power BI dashboards as part of the overall project.



13. MODULE 6 – DEPARTMENT AND RESOURCE ANALYTICS

13.1 Objective

The objective of Module 6 is to complete the remaining dashboard development by retaining the existing Power BI dashboards and adding the remaining Tableau dashboards.

13.2 Power BI Dashboards

Power BI Dashboard 3 and Dashboard 4, developed during Module 4, are retained in the project.

These dashboards form part of the final Power BI dashboard suite.

13.3 Tableau Dashboards

Tableau Dashboard 3 and Dashboard 4 are developed to complete the additional Tableau dashboard analysis.

The dashboards use the final analytical dataset and present the information through visualizations.

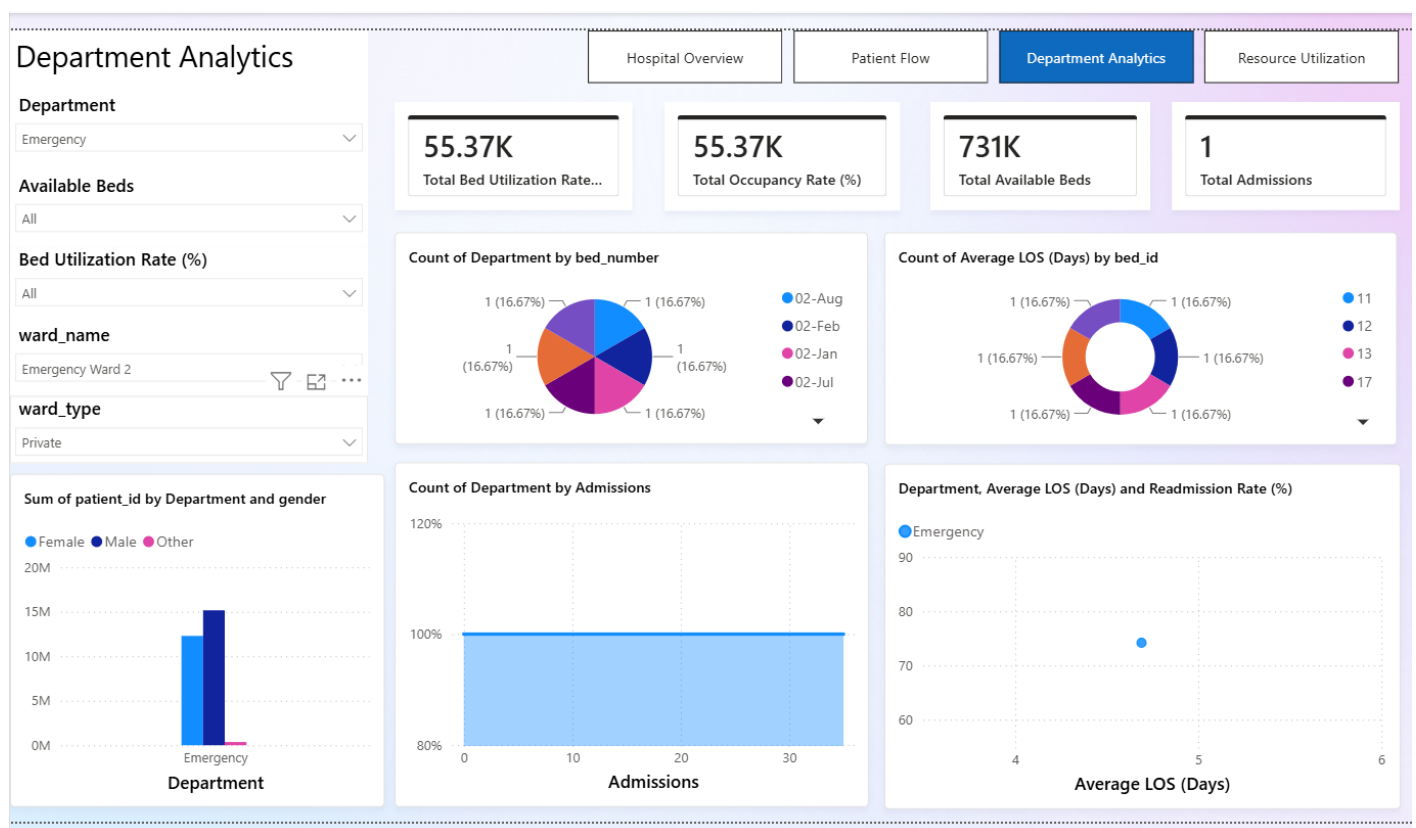
13.4 Department and Resource Analysis

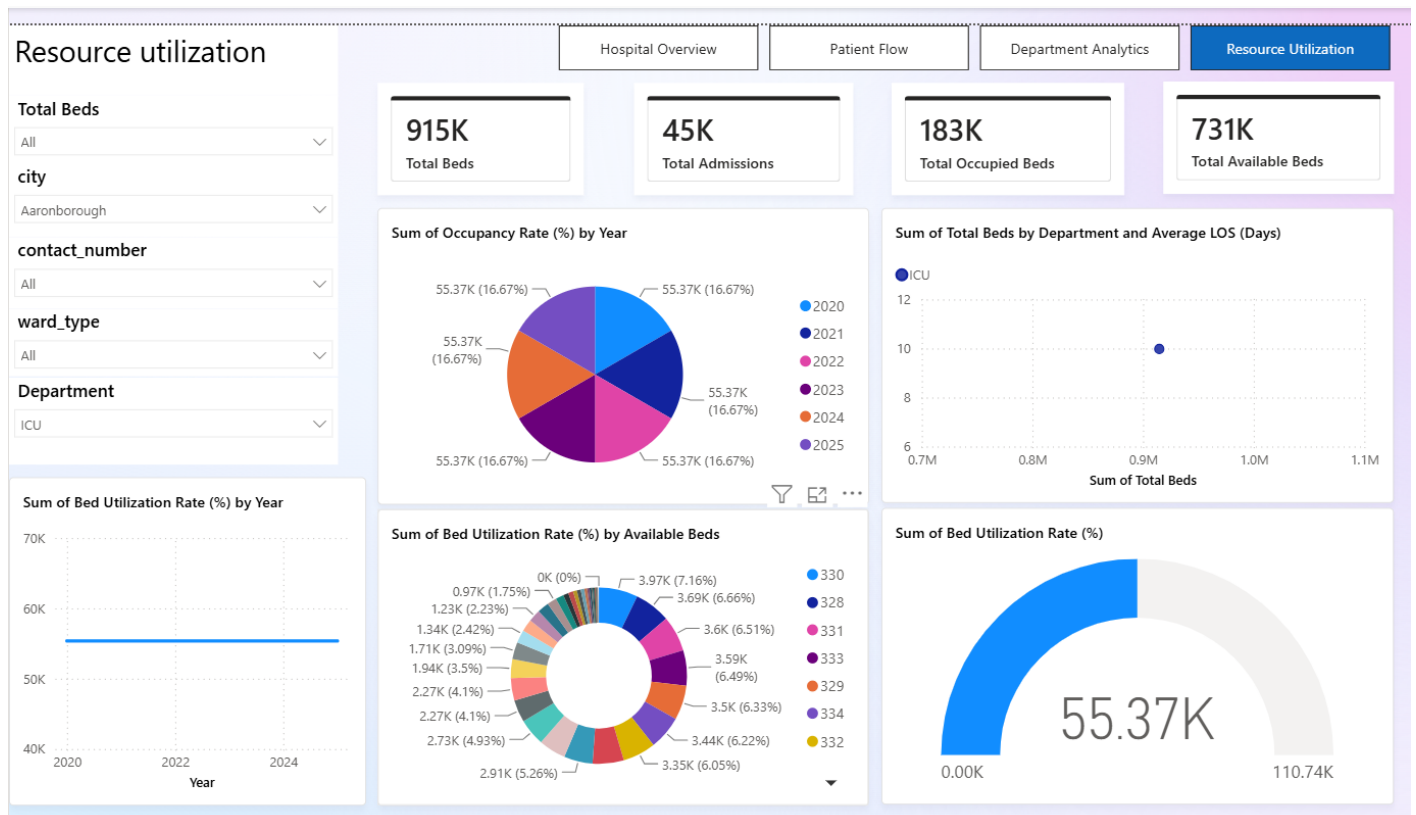
The project plan identifies department analytics and resource utilization as important areas of hospital analysis.

These areas may include department performance, patient volume, bed utilization, resource availability, and capacity-related information, depending on the fields available in the final dataset.

13.5 Result

The remaining Power BI and Tableau dashboards complete the dashboard development process and provide additional views of the hospital analytical data.





14. TABLEAU DASHBOARD ANALYSIS

14.1 Dashboard Overview

The Tableau dashboard suite provides visual representations of the hospital analytical dataset.

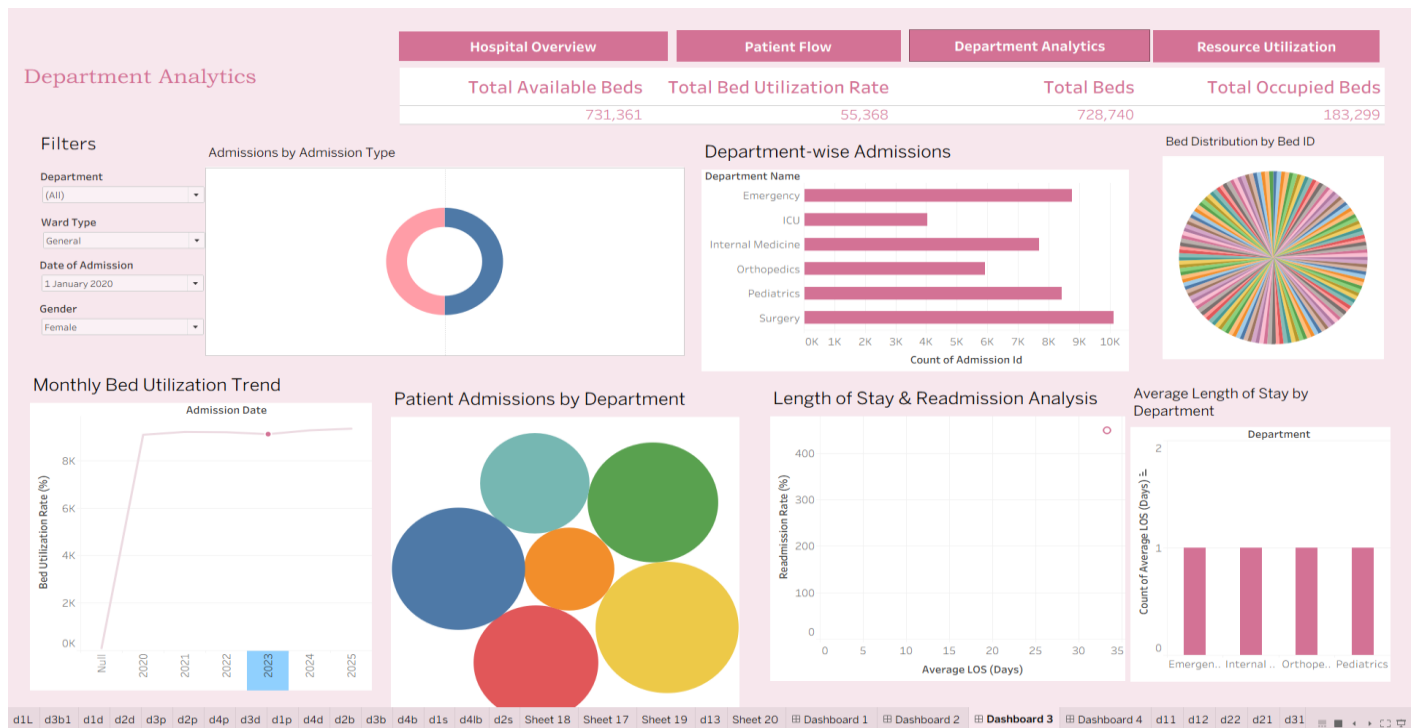
The dashboards are developed to complement the Power BI dashboards and extend the overall project analysis.

14.2 Tableau Dashboard 3

Dashboard title: Insert the exact title shown in Tableau.

This dashboard presents the hospital information displayed in the third Tableau dashboard page.

The visualizations help users explore the available data and understand the analytical information represented.

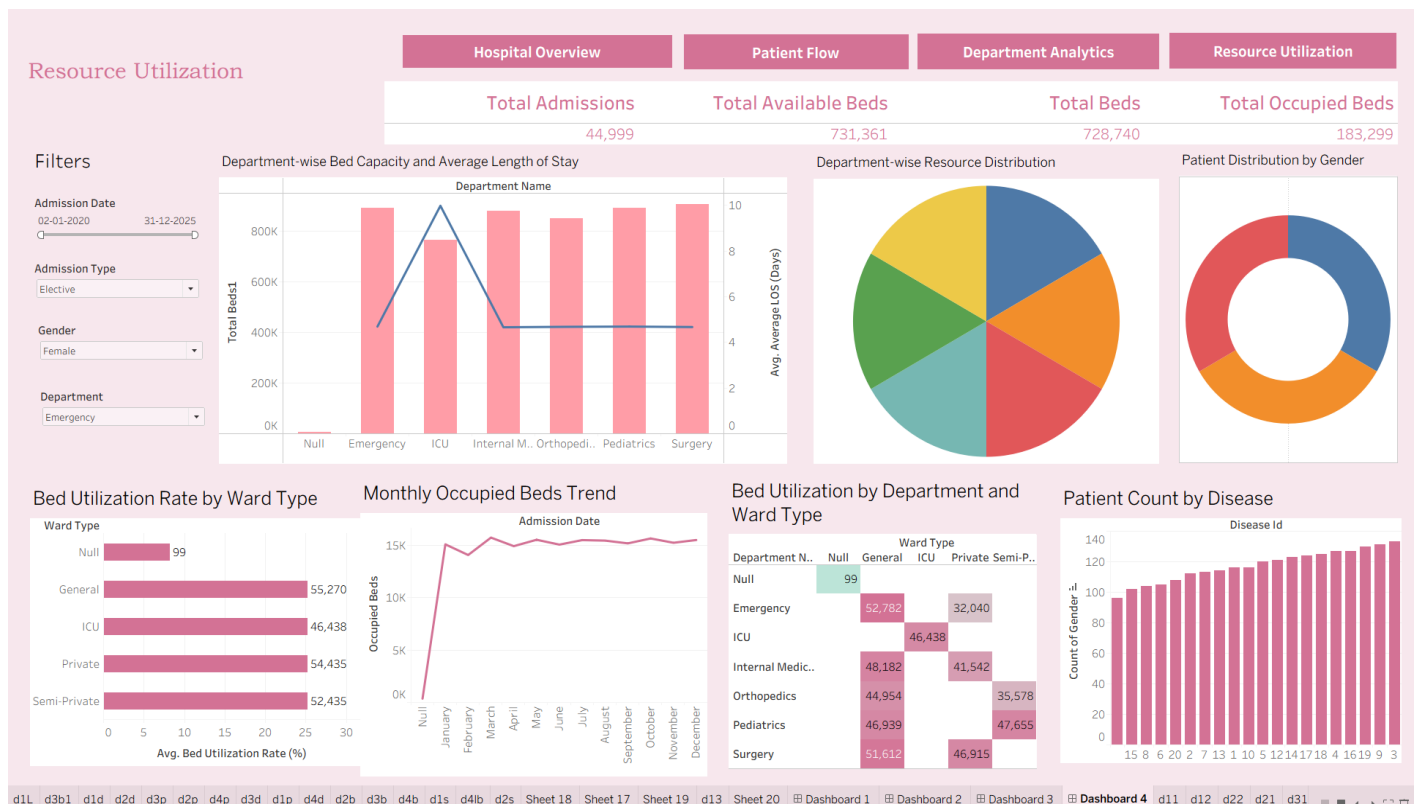


14.3 Tableau Dashboard 4

Dashboard title: Insert the exact title shown in Tableau.

This dashboard completes the additional Tableau dashboard development.

It provides another visual perspective on the hospital analytical data and contributes to the overall dashboard suite.



14.4 Dashboard Interaction

The available filters and interactive features help users explore the displayed information and examine different aspects of the data.

The final dashboard suite combines the visualizations developed using both Power BI and Tableau.

15. MODULE 7 – DASHBOARD INTEGRATION

15.1 Objective

The objective of Module 7 is to organize the completed dashboards into final Power BI and Tableau deliverables.

15.2 Integration Process

The dashboards developed in the previous modules are collected and organized into their respective dashboard files.

The integration process includes:

1. Collecting the completed Power BI dashboards.
2. Collecting the completed Tableau dashboards.
3. Organizing the dashboards into the final files.
4. Reviewing the dashboard pages and available interactions.
5. Preparing the final project deliverables.

15.3 Final Dashboard Structure

The project contains two main dashboard deliverables:

Deliverable	Contents
Power BI Dashboard File	The completed Power BI dashboards
Tableau Dashboard File	The completed Tableau dashboards

15.4 Result

The completed dashboards are organized into separate Power BI and Tableau files.

This integration makes the dashboard suite easier to present, review, and maintain.

16. TESTING AND VALIDATION

16.1 Objective

The objective of testing and validation is to review the processed datasets, KPI calculations, and dashboard functionality.

16.2 Testing Activities

Testing Area	Purpose
Dataset Validation	Review the processed datasets
KPI Validation	Check the generated performance indicators
Visualization Testing	Examine the displayed charts and values
Filter Testing	Verify the available dashboard filters
Navigation Testing	Check the dashboard navigation features
Integration Testing	Review the final dashboard files

16.3 Validation Procedure

1. Review the final analytical dataset.
2. Examine the generated KPI values.
3. Check whether the dashboards display the intended information.
4. Test the available filters and navigation features.
5. Review the final Power BI and Tableau dashboard files.

16.4 Project Outcomes

The project produces a structured analytical dataset and a set of interactive dashboards for hospital data analysis.

The main outcomes include:

- Integration of four hospital datasets.
- Python-based data processing and cleaning.
- Generation of hospital performance indicators.
- Creation of an Excel analytical dataset.
- Development of Power BI and Tableau dashboards.
- Organization of the final dashboard deliverables.

16.5 Applications

The dashboards can support the exploration of hospital admissions, patient-related trends, department performance, and resource utilization using the available dataset.

17. CONCLUSION AND REFERENCES

17.1 Conclusion

MedTrack_DV – Hospital Operations & Patient Analytics Dashboard demonstrates how healthcare data can be transformed into meaningful visual insights through a structured data analytics workflow.

The project begins with the collection of four hospital datasets from Kaggle: admission.csv, bed.csv, patient.csv, and ward.csv.

Python is used to integrate and process the datasets sequentially, generate the required analytical files, and calculate hospital performance indicators.

The final analytical dataset, hospital_final_dataset.xlsx, is then used to develop interactive dashboards using Microsoft Power BI and Tableau.

The completed dashboards provide visual representations of hospital-related information and support the exploration of patient admissions, hospital performance, department analytics, and resource utilization.

Through this project, practical experience is gained in data collection, data processing, KPI generation, dashboard development, and data visualization.

17.2 References

1. Kaggle – Source of the hospital datasets used in the project.
2. HOPA.pdf – Project plan and module-wise implementation procedure.
3. Python Documentation – Reference for Python programming.
4. Pandas Documentation – Reference for data manipulation and processing.
5. Microsoft Power BI – Dashboard development and visualization tool.
6. Tableau – Data visualization and dashboard development tool.

17.3 Final Project Deliverables

- admission.csv
- bed.csv
- patient.csv

- ward.csv
- Hospital patient.csv
- Hospital_raw_dataset.csv
- hospital_cleaned.csv
- hospital_cleaning.ipynb
- generate_hospital_kpis.py
- hospital_final_dataset.xlsx
- Final Power BI dashboard file
- Final Tableau dashboard file